

1. Group Member
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 - b. Louisa Zhang (louisaz@kth.se)
 - c. Sonia Lisboa (lisboa@kth.se)
2. Project Title: Explore Transfer Learning: Fine-Tuning Strategies and Model Depth for Pet Breed Classification
3. Project Type: Default Project 1: Image Classification with Transfer Learning
4. Problem description and approach

This project aims to explore how different transfer learning strategies affect performance on image classification tasks. We use a ResNet-18 pretrained on ImageNet to adapt it for pet breed classification. We do linear probing, fine-tuning l layers simultaneously, and gradual un-freezing to fine tune the network. We will repeat the experiments with different fractions of the dataset, and with imbalanced classes to see how these impact performance. At the end, we will use CutMix to investigate how data-augmentation impacts limited data.
5. Dataset

We will use the Oxford-IIIT Pet Dataset, which contains 7,349 images of cats and dogs across 37 breeds. We will create a training set, validation set and test set.
6. Deep Learning frameworks: PyTorch, Torchvision, matplotlib, numpy, Scikit-learn
7. Implementation plan

The group will write the code from scratch; Implement linear probing, fine-tuning the last l layers, and gradual unfreezing. Repeat this for 100%, 10%, and 1% labeled data. Do the experiment for the imbalance dataset. Implement CutMix augmentation, and compare to the basic augmentation at 100%, 10%, and 1% data. We will refer to official PyTorch and FastAI tutorials to guide the implementation of fine-tuning logic.
8. Baselines
 - a. Linear probing
 - b. Fine-tuning last layer
 - c. Gradual un-freezing
9. Evaluation
 - using:
 - a. Classification accuracy
 - b. Per-class accuracy
 - c. F1-score for imbalance dataset
 - results:
 - a. Different training data sizes
 - b. Different transfer learning strategies
10. Target grade: C/D
11. Milestones
 - a. April 27: Implement linear probing
 - b. May 1: Fine-tune the last l layers
 - c. May 6: Implement gradual unfreezing
 - d. May 11: Run the data on 100%, 10%, 1% data. Run the experiment on an imbalance dataset

- e. May 18: Implement CutMix augmentation
- f. May 23: Analysis and report

12. Learning goals

- a. David
 - i. Understand transfer learning and fine-tuning strategies
 - ii. Learn how model depth and data size affect performance
 - iii. Improve skills in training and evaluating neural network
 - iv. Gain experience using PyTorch
- b. Louisa
 - i. Learn what fine-tuning strategies improve accuracy the most
 - ii. Gain experience with Pytorch and Torchvision
 - iii. Understand how CutMix impacts performance
- c. Sonia: first deep learning course. My goals for completing the project are:
 - i. Learn how pre-trained ConvNet can be adapted to our dataset and gain hands-on experience with transfer learning
 - ii. Understand and compare different adaptation techniques; linear probing & fine tuning
 - iii. Investigate how different factors like model depth, different optimizers and amount of training data affect final test accuracy

13. Workload justification

- a. David
 - i. Set up Pytorch with ResNet-18
 - ii. Load and transform data, create training/test splits
 - iii. Implement Linear probing
- b. Louisa
 - i. Implement imbalanced class experiment
 - ii. Fine-tuning strategies on l layers
 - iii. Experiments on 100%, 10%, 1% data
- c. Sonia
 - i. Implement gradual unfreezing
 - ii. Implement CutMix implementation
 - iii. Write report and include figures